

## LUKE AFB, AZ, USA

WMO: 722785

Lat: 33.533N

Lon: 112.383W

Elev: 331

StdP: 97.41

Time Zone: -7.00 (W07)

Period: 94-19

WBAN: 23111

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|-------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 12            | 1.7        | 3.1 | -14.9                         | 1.1 | 17.8 | -12.6 | 1.3 | 18.5 | 10.3                  | 13.6 | 8.3 | 12.3 | 2.4                   | 340  | 0.461 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 13.2                   | 43.9            | 20.9 | 42.6 | 20.9 | 41.4 | 20.8 | 24.5                | 33.5 | 24.0 | 33.9 | 23.5 | 34.2 | 5.1                  | 220  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 22.6                            | 18.1 | 27.2 | 21.5 | 16.8 | 28.2 | 20.3 | 15.6 | 29.4 | 75.5          | 33.3 | 73.4 | 33.4 | 71.6 | 34.0 | 28.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |     |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|-----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |     |
| (4) 9.3           | 8.1  | 7.1 | DB                         | -0.9 | 46.6               | 1.8 | 1.5                                                | -2.1 | 47.6       | -3.2 | 48.5       | -4.2 | 49.3       | -5.4 | 50.4 | (4) |
| (5)               |      |     | WB                         | -3.2 | 25.8               | 2.0 | 0.9                                                | -4.6 | 26.4       | -5.7 | 26.9       | -6.8 | 27.4       | -8.2 | 28.1 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 22.8        | 12.2  | 14.1  | 17.9  | 21.3  | 25.7  | 31.4  | 34.4  | 33.9  | 30.3  | 23.2  | 17.0  | 11.8  | (6)  |  |
|                                                                        | DBStd       | 8.58        | 2.86  | 3.39  | 3.17  | 3.59  | 3.47  | 3.09  | 2.33  | 2.31  | 2.77  | 3.48  | 3.33  | 2.95  | (7)  |  |
|                                                                        | HDD10.0     | 34          | 12    | 6     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 16    | (8)  |  |
|                                                                        | HDD18.3     | 643         | 189   | 122   | 47    | 13    | 1     | 0     | 0     | 0     | 0     | 6     | 63    | 202   | (9)  |  |
|                                                                        | CDD10.0     | 4716        | 82    | 119   | 246   | 338   | 487   | 642   | 758   | 742   | 609   | 410   | 210   | 72    | (10) |  |
|                                                                        | CDD18.3     | 2282        | 0     | 3     | 34    | 102   | 229   | 392   | 499   | 484   | 359   | 157   | 22    | 0     | (11) |  |
|                                                                        | CDH23.3     | 36040       | 14    | 97    | 587   | 1427  | 3304  | 6591  | 8492  | 8066  | 5261  | 1871  | 323   | 8     | (12) |  |
|                                                                        | CDH26.7     | 23242       | 0     | 10    | 186   | 641   | 1879  | 4522  | 6071  | 5656  | 3283  | 914   | 80    | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 3.0         | 2.5   | 2.7   | 2.9   | 3.5   | 3.5   | 3.5   | 3.7   | 3.4   | 3.0   | 2.7   | 2.5   | 2.3   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 210         | 25    | 26    | 22    | 7     | 3     | 1     | 20    | 32    | 19    | 14    | 14    | 27    | (15) |  |
|                                                                        | PrecMax     | 451         | 167   | 117   | 79    | 40    | 32    | 10    | 64    | 73    | 73    | 95    | 64    | 102   | (16) |  |
|                                                                        | PrecMin     | 106         | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 4     | 0     | 1     | 0     | 1     | (17) |  |
|                                                                        | PrecStd     | 83          | 37    | 33    | 19    | 9     | 8     | 3     | 17    | 18    | 19    | 18    | 15    | 25    | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 25.2  | 28.2  | 33.0  | 36.7  | 40.7  | 46.1  | 45.9  | 44.9  | 42.0  | 37.6  | 31.4  | 24.5  | (19) |  |
|                                                                        |             | MCWB        | 11.4  | 13.1  | 14.7  | 16.0  | 17.6  | 20.7  | 21.2  | 21.4  | 21.2  | 18.8  | 14.8  | 12.3  | (20) |  |
|                                                                        | 2%          | DB          | 22.9  | 26.2  | 30.4  | 34.5  | 38.5  | 43.2  | 43.9  | 43.2  | 40.5  | 35.2  | 28.9  | 22.4  | (21) |  |
|                                                                        |             | MCWB        | 10.9  | 12.7  | 13.8  | 15.3  | 16.7  | 19.3  | 21.1  | 21.9  | 20.5  | 17.3  | 14.0  | 11.5  | (22) |  |
|                                                                        | 5%          | DB          | 21.2  | 24.2  | 28.8  | 32.6  | 36.8  | 41.9  | 42.7  | 42.1  | 39.1  | 33.6  | 27.1  | 20.6  | (23) |  |
|                                                                        |             | MCWB        | 10.4  | 12.2  | 13.2  | 14.5  | 16.4  | 18.9  | 21.3  | 21.8  | 20.2  | 16.6  | 13.6  | 10.8  | (24) |  |
|                                                                        | 10%         | DB          | 19.3  | 22.2  | 26.9  | 30.6  | 35.1  | 40.4  | 41.4  | 40.8  | 37.8  | 31.9  | 25.0  | 18.6  | (25) |  |
|                                                                        |             | MCWB        | 9.8   | 11.4  | 12.7  | 13.8  | 15.9  | 18.4  | 21.3  | 21.9  | 20.4  | 16.2  | 13.0  | 10.0  | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 15.2  | 15.5  | 16.0  | 17.3  | 19.1  | 22.9  | 25.2  | 25.3  | 24.6  | 22.1  | 16.7  | 15.2  | (27) |  |
|                                                                        |             | MCDB        | 18.0  | 22.6  | 27.8  | 33.1  | 32.1  | 40.1  | 31.5  | 34.3  | 30.6  | 28.1  | 24.2  | 18.7  | (28) |  |
|                                                                        | 2%          | WB          | 13.4  | 14.3  | 15.1  | 16.0  | 18.0  | 21.7  | 24.4  | 24.6  | 23.8  | 19.9  | 15.6  | 13.9  | (29) |  |
|                                                                        |             | MCDB        | 17.1  | 21.6  | 26.0  | 31.8  | 34.2  | 38.1  | 33.2  | 34.4  | 31.9  | 29.9  | 24.7  | 18.4  | (30) |  |
|                                                                        | 5%          | WB          | 12.1  | 13.4  | 14.2  | 15.2  | 17.3  | 20.6  | 23.9  | 24.1  | 23.1  | 18.7  | 14.7  | 12.8  | (31) |  |
|                                                                        |             | MCDB        | 17.4  | 20.8  | 24.7  | 30.1  | 34.1  | 37.6  | 33.8  | 34.7  | 32.6  | 29.0  | 24.3  | 17.3  | (32) |  |
|                                                                        | 10%         | WB          | 10.9  | 12.4  | 13.5  | 14.4  | 16.4  | 19.5  | 23.5  | 23.6  | 22.5  | 17.6  | 13.8  | 11.6  | (33) |  |
|                                                                        |             | MCDB        | 17.1  | 20.1  | 24.3  | 29.0  | 33.2  | 37.8  | 34.3  | 34.8  | 32.5  | 28.7  | 23.5  | 16.7  | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 13.4  | 13.8  | 15.4  | 15.5  | 15.4  | 16.4  | 13.2  | 12.8  | 13.6  | 15.2  | 14.6  | 12.8  | (35) |  |
|                                                                        |             | MCDBR       | 16.3  | 17.0  | 18.2  | 18.3  | 17.8  | 17.5  | 15.0  | 14.5  | 15.2  | 17.1  | 17.0  | 15.4  | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 7.5   | 7.3   | 6.5   | 6.5   | 5.7   | 5.4   | 4.2   | 3.8   | 4.4   | 6.3   | 7.2   | 7.3   | (37) |  |
|                                                                        |             | MCWBR       | 11.7  | 14.2  | 15.8  | 16.3  | 15.8  | 14.8  | 11.7  | 11.9  | 11.2  | 13.3  | 14.4  | 11.6  | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.281 | 0.286 | 0.311 | 0.325 | 0.335 | 0.344 | 0.403 | 0.392 | 0.348 | 0.319 | 0.289 | 0.280 | (40) |  |
|                                                                        |             | taud        | 2.567 | 2.561 | 2.468 | 2.412 | 2.413 | 2.389 | 2.351 | 2.375 | 2.468 | 2.505 | 2.547 | 2.571 | (41) |  |
|                                                                        | Ebn at Noon |             | 948   | 976   | 968   | 962   | 948   | 935   | 878   | 885   | 915   | 922   | 927   | 927   | (42) |  |
|                                                                        |             | Edh at Noon | 79    | 88    | 105   | 116   | 118   | 120   | 124   | 120   | 104   | 92    | 80    | 75    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 3.39  | 4.36  | 5.84  | 7.26  | 8.14  | 8.44  | 7.43  | 6.82  | 6.13  | 4.92  | 3.75  | 3.07  | (44) |  |
|                                                                        | RadStd      |             | 0.24  | 0.32  | 0.27  | 0.20  | 0.25  | 0.30  | 0.33  | 0.22  | 0.21  | 0.27  | 0.15  | 0.16  | (45) |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (13 neighbors) |  | +0.34 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -83     | +107    | +38     |

Nomenclature: See separate page